# Exercise-7

## Problem Statement: Financial Forecasting

## Source code:

```
package dsa;

import java.util.Scanner;

public class FinancialForecaster {

    public static double calculateFutureValue(double presentValue, double growthRate, int
periods) {

        if (periods == 0) {

            return presentValue;

        }

        double nextValue = presentValue * (1 + growthRate);

        return calculateFutureValue(nextValue, growthRate, periods - 1);

    }

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter initial investment amount: ");

        double initialInvestment = scanner.nextDouble();

        System.out.print("Enter annual growth rate (e.g., 0.05 for 5%): ");

        double annualGrowth = scanner.nextDouble();

        System.out.print("Enter number of years to forecast: ");

        int years = scanner.nextInt();

        double forecastedValue = calculateFutureValue(initialInvestment, annualGrowth, years);

        System.out.printf("Initial Investment: $%.2f%n", initialInvestment);

        System.out.printf("Forecasted Value after %d years: $%.2f%n", years, forecastedValue);

        scanner.close();

    }
```

}

**Output**:

```
Console
<terminated> FinancialForecaster [Java Application] C:\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_17.0.3.v20220515-1416\jre\bin\javaw.exe
Enter initial investment amount: 45000
Enter annual growth rate (e.g., 0.05 for 5%): 0.2
Enter number of years to forecast: 3
Initial Investment: $45000.00
Forecasted Value after 3 years: $77760.00
```